Bare-Metal Provisioning MAAS API

❖ **The following sequence dictates the order of development, ensuring that each API build rests upon a stable, functional foundation.**

Priority	API Name	Primary Purpose	Prerequisite/Dependency	Example API Endpoint
1	Machine Discovery	Real-time inventory visibility of bare-metal servers	None	GET /machines
2	Machine Details	Retrieve comprehensive hardware specifications & status	Discovery API	GET /machines/{id}
3	Machine Commissioning	Baseline hardware validation and MAAS enrollment	Details API	POST /machines/{id}/commission
4	Machine Allocation	Reserve a machine for specific deployment/user	Commissioning API	POST /machines/{id}/allocate
5	Storage/RAID Config	Define logical disk topology (RAID levels, partitions)	Allocation API	PUT /machines/{id}/storage
6	Provisioning Payload	Specify OS, user accounts, SSH keys, and cloud-init	Storage/RAID Config	PUT /machines/{id}/payload
7	OS Deployment	Initiate the automated operating system installation	Provisioning Payload	POST /machines/{id}/deploy
8	Deployment Status	Monitor real-time progress and logs of deployment	OS Deployment API	GET /machines/{id}/deployment/status
9	Power Management	Remote control: power on/off, reboot, cycle	Core Deployment Flow	POST /machines/{id}/power
10	Release/Deallocate	Return machine to available pool; optional data wipe	Deployment Completion	POST /machines/{id}/release
11	Reinstallation	Re-deploy OS with fresh parameters/image	Established Lifecycle	POST /machines/{id}/reinstall
12	Tagging & Metadata	Categorize machines for management and automation	Established Lifecycle	PUT /machines/{id}/tags
13	Network Configuration	Configure VLANs, bonding, subnets, IP assignments	OS Deployment	PUT /machines/{id}/network
14	Event & Audit Logs	Trace all actions for compliance and debugging	Any API Action	GET /machines/{id}/audits
15	Access Control (RBAC)	Define user roles and permissions for API access	All Core APIs	PUT /users/{user_id}/permissions

Storage & RAID Orchestration: The Storage/RAID API allows the system to define the logical disk topology before the operating system is installed. This ensures that performance and data redundancy requirements (RAID 0, 1, 5, 10) are enforced programmatically.

Provisioning Payload (OS & Authentication): The Provisioning Payload API encapsulates the identity and configuration of the host. This endpoint will accept specific parameters to ensure "Zero-Touch" deployment: 1. Distribution & Version: Target OS (e.g., Ubuntu 24.04 LTS). 2. Authentication & Access: Administrative username, password hashing, and injection of authorized SSH public keys. 3. Customization: Hooks for cloud-init directives and post-installation configuration scripts.